Supplementary Reproducibility Statement

Constraint Audited Receding Horizon Dispatch for Curtailment Recovery in Export Capped Utility Scale Photovoltaic Plants with Battery Energy Storage

1. Study and package scope

The research workflow comprises twelve chronological monthly folds (September 2024–August 2025), a 3 h dispatch interval, matched History-CatBoost and Weather-CatBoost forecasts, a Diurnal reference, and a finite-horizon perfect-PV information comparator. Both 24 h and 72 h mixed-integer receding-horizon controllers use the same BESS and POI constraints. The common terminal treatment consists of 168 h of preparation and a 24 h endgame, with a 50% stored-energy target at each monthly boundary. The materials also document HGB model-family robustness, independently varied BESS power and energy, and paired monthly statistical inference.

2. Executable source and order of operations

The public archive provides a single nominal-control entry point, run_nominal_control.py, with the matched forecast table as its required input. It calls the stored forecasting-to-control mapping and the shared-terminal MILP–MPC implementation. The underlying source modules, published configurations, model-selection records, aggregate evidence tables, and command-line validation utilities are supplied. Forecast model fitting and HGB/BESS studies have separate source scripts; their complete reproduction requires the private supervised records and frozen forecasts documented in the input specification. The reference control settings are preserved without new optimization or refitting in the present package.

3. Restricted inputs and reproduction boundary

Raw hourly plant active-power records are held under contractual confidentiality and are not distributed. The complete leadwise supervised table, full time-resolved forecast tables, and high-frequency closed-loop trajectories are likewise withheld because the true PV targets and perfect-information sequences can disclose the underlying plant series. Exact reproduction of numerical plant-specific outcomes requires authorized access to these private inputs. A sanitized table, column dictionary, forecast-content authentication manifest, and structured input description document the expected interface. Public aggregate checks authenticate the recorded campaign values without reconstructing private observations.

4. Source environment and solver

The recorded software environment used Python 3.13.5, CatBoost 1.2.8, NumPy 2.3.5, pandas 2.2.3, scikit-learn 1.8.0, and SciPy 1.17.0 with the HiGHS backend via scipy.optimize.milp. The relative MIP-gap option was 1e-8. Across 23,336 optimal MILP calls, the mean, median, 95th-percentile and maximum solve times were 1.449, 1.415, 2.227 and 41.118 ms; the maximum observed relative gap was 9.96e-11 and maximum branch-and-bound nodes one. These timings were obtained on Linux (kernel 6.12.13, glibc 2.41) with an AMD EPYC 9V74 processor, four allocated CPU cores and a 4 GiB memory limit. The nominal 24 h and 72 h MILPs contain 43 and 123 decision variables, including 8 and 24 binary variables. Timings exclude forecast generation, data transfer, supervisory execution and plant protection.

5. Forecast provenance, quantitative evidence and checksums

The frozen nominal forecast manifest authenticates 266,784 forecast-table records by compressed-file and logical-content SHA-256 digests. Control comparisons use the common terminal-steering implementation throughout the final 192 h of each monthly fold. The public package includes the nominal control aggregates, HGB robustness summaries, paired monthly bootstrap statistics, model-selection records, and an SHA-256 index covering each distributed file. The accompanying validation utility checks checksums, configuration consistency and reported annual KPI references. These are evidence-integrity checks; they do not replace a private-input experiment rerun.

6. Associated materials

Supplementary Material 1 contains the forecasting, optimization, terminal-treatment, and statistical settings. Supplementary Material 3 provides code, public derived data and authentication files. Supplementary Material 4 presents campaign forecast diagnostics, the monthly weather-information effect, BESS sensitivity, and

paired-interval results.